

Reg No.: _____

0520MCA188052301

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Second Semester MCA (Two Years) Degree Supplementary Examination January 2024

Course Code: 20MCA188

Course Name: ARTIFICIAL INTELLIGENCE

Max. Marks: 60

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

- | | | |
|----|--|-----|
| 1 | What is a Production System in AI? How Production Systems are classified? | (3) |
| 2 | Define 8 Puzzle Problem. Represent 8 puzzle problem using a search tree. | (3) |
| 3 | What are the problems associated with Hill climbing search procedure? | (3) |
| 4 | What is Simulated annealing in AI? | (3) |
| 5 | Distinguish between Semantic network based and frame based knowledge representation schemes with the help of an example. | (3) |
| 6 | Give the conceptual dependency representations of the following sentences:
(a) Bob shot Mary.
(b) John fertilized the field.
(c) Susan drove her car to the office. | (3) |
| 7 | What is a WFF? Explain quantifiers in FOPL with examples. | (3) |
| 8 | Discuss the importance of planning in AI? Explain Hierarchical Planning? | (3) |
| 9 | What is Fuzzy Logic? What are its applications? | (3) |
| 10 | How Expert systems are different from Conventional Programs? Explain. | (3) |

PART B

Answer any one question from each module. Each question carries 6 marks.

Module I

- | | | |
|----|---|-----|
| 11 | How Missionaries and Cannibals problem can be solved using State space representation scheme? | (6) |
|----|---|-----|

OR

- | | | |
|----|---|-----|
| 12 | Explain various AI problem characteristics with appropriate examples. | (6) |
|----|---|-----|

Module II

- | | | |
|----|---|-----|
| 13 | Compare Hill climbing and Steepest ascent hill climbing algorithms with appropriate examples. | (6) |
|----|---|-----|

OR

- 14 Explain various blind search strategies in AI (6)

Module III

- 15 Explain MINMAX algorithm with the help of an example. (6)

OR

- 16 What is Conceptual dependency Theory? What are its components? (6)

Module IV

- 17 John likes all kinds of fruits. Orange is a fruit. Prove that John likes Orange in two ways. (6)

OR

- 18 Explain various learning techniques in AI? What methods are used for learning? (6)

Module V

- 19 What is an Expert system? Explain architecture of an expert system with a diagram. (6)

OR

- 20 Consider the following two fuzzy sets. (6)

$$\tilde{A} = \frac{0.5}{2} + \frac{0.7}{3} + \frac{1}{5}$$

$$\tilde{B} = \frac{0.2}{2} + \frac{0.8}{5}$$

Perform the operations i) Algebraic Sum ii) Bounded sum iii) Bounded Difference.
